

PROJECT DOCUMENTATION

Streamlining IT Procurement: Automatic Standard Laptop Orders With Flow Designer

Project Title	Streamlining IT Procurement: Automatic Standard Laptop Orders With Flow Designer
Technology / Platform	ServiceNow – Flow Designer
Project Type	IT Procurement Automation
Repository	Public GitHub Repository – Streamlining-IT-Procurement
Student Name	Anandhi,Sifana,Priyadharshini,Harshitha Devi,Priyanga Sangari
Institution / Department	St. Joseph's Arts and Science College for Women
Programme / Year	B.Sc. Computer Science – 3rd Year
Submission Date	25.09.26

1. Project Overview

This project demonstrates the automation of a standard laptop procurement process using ServiceNow Flow Designer. The objective is to reduce manual effort in IT procurement by creating a flow that receives a laptop order/request, validates the required information, creates or updates the relevant procurement/request records, and automatically performs the configured downstream actions.

1.1 Problem Statement

Traditional IT laptop procurement may involve manual request handling, repeated data entry, approval follow-up, and communication between employees, IT teams and procurement teams. This can create delays and inconsistent processing. The proposed solution automates a standard laptop-order workflow using Flow Designer.

1.2 Project Objectives

- Create a standardized process for ordering a standard laptop.
- Automate repetitive procurement activities using ServiceNow Flow Designer.
- Reduce manual data entry and improve process consistency.
- Use conditions and actions to control the workflow.
- Maintain request/order information in ServiceNow records.
- Test the flow with valid and invalid scenarios.
- Document the implementation and demonstrate the completed automation.

1.3 High-Level Workflow

Service Catalog Trigger → Standard laptop task flow → Create Catalog Task → Catalog Task record generated in ServiceNow

Implementation evidence in this document is based on the completed ServiceNow configuration shown in the submitted screenshots.

2. Brainstorming & Ideation Phase

Identify the procurement problem, users involved, automation opportunities and the basic solution idea.

Activities / Work to Complete

- Identify the manual steps involved in requesting and processing a standard laptop.
- Identify stakeholders such as employee/requester, IT support, procurement team and approver.
- List repetitive activities that can be automated.
- Discuss what information must be captured for a standard laptop order.
- Decide the automation approach using ServiceNow Flow Designer.
- Define the expected result: a standard laptop order should move through the configured process automatically.

Expected Deliverables

- Problem statement.
- Initial solution idea.
- List of stakeholders.
- List of inputs and expected outputs.
- Simple workflow sketch.

Evidence / Submission Reference

- Screenshot/photo of brainstorming notes.
- Initial workflow diagram.
- GitHub folder: 01_Brainstorming_Ideation/

3. Requirement Analysis Phase

Convert the project idea into functional and non-functional requirements.

Activities / Work to Complete

- Identify the required ServiceNow application/module and Flow Designer access.
- Identify the request/order information required for processing.
- Define the flow trigger based on the project implementation.
- Define conditions for a standard laptop request.
- Define actions that should happen after the condition is satisfied.
- Define notification or record-update requirements if used in the implementation.
- Define the expected successful and unsuccessful outcomes.

Expected Deliverables

- Functional requirements.
- Non-functional requirements.
- Input/output specification.
- User/stakeholder requirements.
- Acceptance criteria.

Evidence / Submission Reference

- Requirements table.
- Screenshot of relevant ServiceNow configuration.
- GitHub folder: 02_Requirement_Analysis/

3.1 Suggested Requirements Table

ID	Requirement	Type	Acceptance Criteria
FR-01	System should receive a standard laptop request/order.	Functional	Request data is captured successfully.
FR-02	Flow should start when the configured trigger occurs.	Functional	Flow execution is initiated automatically.
FR-03	Flow should perform the configured Service Catalog action for the standard laptop task.	Functional	Create Catalog Task action is executed as configured.
FR-04	Flow should create the required Catalog Task record.	Functional	Expected Catalog Task record is generated in ServiceNow.
FR-05	Flow should perform configured notification/action steps.	Functional	Configured action executes successfully.
NFR-01	Process should be consistent and repeatable.	Non-functional	Same valid input produces the expected result.

4. Project Design Phase

Design the architecture, data flow and Flow Designer logic before implementation.

Activities / Work to Complete

- Design the overall system architecture.
- Identify the source/request record and target record(s).
- Design the Flow Designer trigger.
- Design decision/condition logic for the standard laptop.
- Design actions such as record creation/update and notifications, according to the reference implementation.
- Define error/exception handling where applicable.
- Prepare a flowchart or activity diagram.

Expected Deliverables

- System architecture diagram.
- Flowchart.
- Data/input-output mapping.
- Flow logic description.

Evidence / Submission Reference

- Architecture/flow diagram.
- Flow Designer screenshots.
- GitHub folder: 03_Project_Design/

4.1 Proposed Flow Logic

1. Start when the configured laptop-order trigger occurs.
2. Read the relevant request/order information.
3. Check whether the request meets the standard-laptop criteria.
4. If the condition is satisfied, execute the configured procurement action(s).
5. Create or update the required ServiceNow record(s).
6. Send the configured notification or status update, if included.
7. End the flow successfully.
8. If the condition is not satisfied, follow the alternative path configured in the project.

5. Project Planning Phase

Plan the work, phase-wise submission, responsibilities, resources and timeline.

Activities / Work to Complete

- Divide the project into the eight required phases.
- Create a GitHub repository and make it Public.
- Create folders for each phase.
- Plan the ServiceNow implementation work.
- Plan testing scenarios and evidence collection.
- Plan final documentation and demonstration.
- Maintain meaningful GitHub commits for each completed phase.

Expected Deliverables

- Project schedule.
- GitHub repository structure.
- Task checklist.
- Milestone/commit plan.

Evidence / Submission Reference

- Public GitHub repository screenshot.
- Repository folder structure screenshot.
- GitHub commit history screenshot.
- GitHub folder: 04_Project_Planning/

5.1 Recommended GitHub Structure

Streamlining-IT-Procurement/

```
|
|— 01_Brainstorming_Ideation/
|— 02_Requirement_Analysis/
|— 03_Project_Design/
|— 04_Project_Planning/
|— 05_Project_Development/
|— 06_Project_Testing/
|— 07_Project_Documentation/
|— 08_Project_Demonstration/
|— README.md
```

6. Project Development Phase

The project was implemented in ServiceNow Workflow Studio / Flow Designer using the completed configuration shown in the screenshots.

Activities / Work to Complete

- Accessed the assigned ServiceNow developer instance.
- Opened Workflow Studio / Flow Designer and the project flow.
- Created the flow named “Standard laptop task”.
- Configured the trigger as “Service Catalog”.
- Used the configured Service Catalog flow structure shown in the implementation.
- Added the “Create Catalog Task” action.
- Configured the flow to create a Catalog Task record in ServiceNow.
- The flow is shown as Active, with Status: Published and Application: Global.
- Verified the resulting Catalog Task record created by the flow.

Expected Deliverables

- Completed flow: “Standard laptop task”.
- Trigger: Service Catalog; Action: Create Catalog Task.
- A Catalog Task record was generated and displayed as SCTASK0010001.
- ServiceNow implementation screenshots are included below as evidence.

Evidence / Submission Reference

- Flow Designer overview screenshot.

- Trigger screenshot.
- Screenshot showing the Service Catalog trigger and Create Catalog Task action.
- Screenshot of the resulting Catalog Task record.
- GitHub folder: 05_Project_Development/

6.1 Development Evidence Checklist

- ☒ Flow name visible: Standard laptop task.
- ☒ Trigger configured: Service Catalog.
- ☒ Configured flow structure shown in Workflow Studio.
- ☒ Action configured: Create Catalog Task.
- ☒ Flow is shown as Active / Published.
- ☒ Flow is shown as Active.
- ☒ Resulting Catalog Task record is available as evidence.
- ☒ Catalog Task SCTASK0010001 is visible with Approved approval and Open state.

Implementation Evidence Screenshot

The screenshot displays the ServiceNow Workflow Studio interface for a flow named "Standard laptop tasks". The flow is currently in an "Inactive" state. The trigger is configured as "Service Catalog". The actions section is empty, with a button to "Add an Action, Flow Logic, or Subflow". The error handler is disabled. The data panel on the right shows the "Trigger - Service Catalog" configuration, including "Requested Item Record", "Run Start Time UTC", "Run Start Date/Time", and "Table Name".

Workflow Studio Interface:

- Flow Name:** Standard laptop tasks (Flow - Global)
- Status:** Inactive
- Trigger:** Service Catalog
- Advanced Options:** (Dropdown menu)
- Buttons:** Delete, Cancel, Done
- Actions:** Select multiple (Add an Action, Flow Logic, or Subflow)
- Error Handler:** (Disabled)
- Data Panel:**
 - Flow Variables
 - Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Run Start Date/Time (Date/Time)
 - Table Name (Table Name)

System Information:

- Status: Draft | Application: Global
- Weather: 32°C Mostly cloudy
- Search: Search
- Language: ENG IN
- Time: 18:07 24/09/2026

Workflow Studio

Standard laptop tasks

Flow • Global

Action: Create Catalog Task

Table Name: Catalog Task [sc_task]

* Requested Item [Requested Item]: Trigger -> Requested Item ...

Short Description: Laptop need to Configured

Fields:

- Description: Laptop need to Configured
- Assignment group: Hardware
- Approval: Approved

+ Add field value

Wait: ☒

Template Catalog Item [Catalog Item]: Select Template Catalog Item

Status: Modified | Application: Global

32°C Mostly cloudy

Search

ENG IN

18:20 24/09/2026

Data

Collapse All

Flow Variables

Trigger - Service Catalog

- Requested Item Record: Record
- Run Start Time UTC: DateTime
- Run Start Date/Time: DateTime
- Table Name: Table Name

1 - Create Catalog Task

- Catalog Task: Record
- Action Status: Object

Standard Laptop | Catalog Item

Standard Laptop

Build and modify items faster with the improved Catalog Builder.

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Standard Laptop

Application: Global

Catalogs: Service Catalog

Category: Hardware

State: -- None --

Checked out: -- None --

Owner:

Active: ☒

Fulfillment automation level: Unspecified

Item Details | Process Engine | Picture | Pricing | Portal Settings

Select the appropriate process engine for the catalog item. Only one engine can be selected.

33°C Mostly cloudy

Search

ENG IN

18:34 24/09/2026

(1) WhatsAppCatalog | ServiceNowStandard laptop tasks | WorkfloAsk Gemini

dev434250.service-now.com/now/nav/ui/classic/params/target/catalog_home.do%3Fsysparm_view%3Dcatalog_default

servicenowAllFavoritesHistoryWorkspacesCatalogSearch

Service CatalogSearch catalog

Services



Services

Document production services. Create and produce high-quality, professional documents.

Can We Help You?



Can We Help You?

Your IT gateway. Report issues and submit requests.

Office



Office

Office services such as printing, supplies requisition and document shipping and delivery.

Peripherals



Peripherals

End user peripherals such as mobile phone cases, dongles, and cables

Hardware



Hardware

Order from a variety of hardware to meet your business needs, including phones, tablets and laptops.

Software



Software

A range of software products available for installation on your corporate laptop or desktop computer.

Desktops



Desktops

Desktop computers for your work area.

Mobiles



Mobiles

Cell phones to meet your business needs.

Top Requests



Request email alias



Access



Cisco jabber softphone



Standard Laptop



Pixel 4a

Shopping Cart

Empty

Hot days ahead

33°C

Search

ENG IN

18:37

24/09/2026

(1) WhatsAppHardware | ServiceNowStandard laptop tasks | WorkfloAsk Gemini

dev434250.service-now.com/now/nav/ui/classic/params/target/com.glideapp.servicecatalog_category_view.do%3Fv%3D1%26sysparm_parent%3Dd258b953c61122...

servicenowAllFavoritesHistoryWorkspacesHardwareSearch



Key Features:

- Screen size: 10.2 inch
- Operating system: iPadOS

if so combines the power and capability of a computer with the ease of use and versatility you'd never expect from one. It has a larger 10.2-inch retina display, support for the full-size smart keyboard and the amazing new capabilities of iPadOS

iPad pro

Request for iPad pro

Preview

\$799.00

Sales Laptop

Acer Aspire NX

Preview

\$1,100.00

Standard Laptop

Lenovo - Carbon x1

Preview

\$1,100.00

Apple Watch

Apple Watch - Their most personal device ever

Preview

\$349.99

Apple MacBook Pro 15"

\$1,099.99

33°C

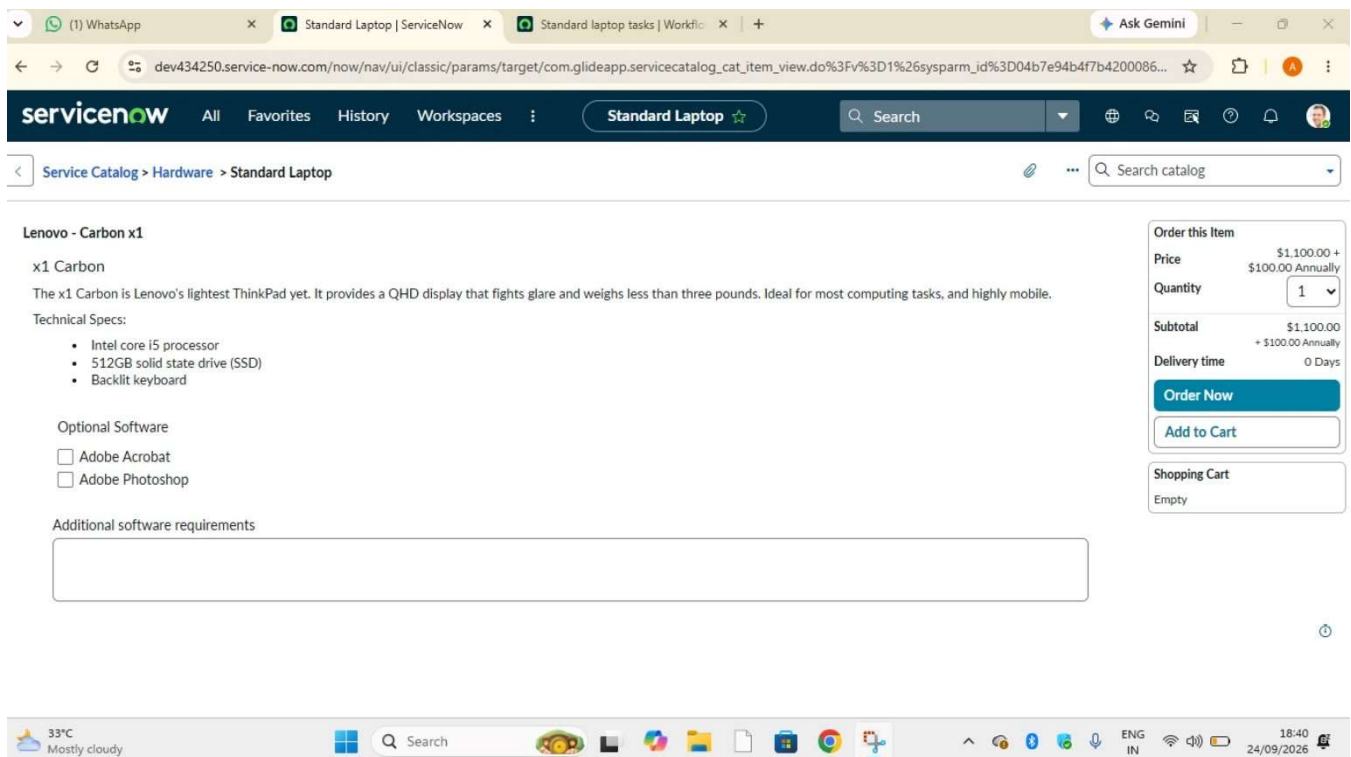
Mostly cloudy

Search

ENG IN

18:39

24/09/2026



6.2 Actual Implementation Details

The implemented flow is named “Standard laptop task”. It uses the Service Catalog trigger and the Create Catalog Task action. The flow is shown as Active and Published in Workflow Studio. The resulting Catalog Task record is SCTASK0010001, with Approval = Approved, State = Open, Request Item = RITM0010001, and Requested For = System Administrator.

7. Project Testing Phase

Testing was performed by verifying the configured ServiceNow flow and the resulting Catalog Task record shown in the submitted screenshots.

Activities / Work to Complete

- Verified the “Standard laptop task” flow and its Service Catalog trigger.
- Verified the resulting Catalog Task record generated in ServiceNow.
- Confirmed that the “Create Catalog Task” action resulted in Catalog Task SCTASK0010001.
- Verified the resulting task status shown in the record: Approved approval and Open state.
- Recorded the observed result in the test table below.
- No additional defect/fix evidence was provided in the screenshots.

Expected Deliverables

- Test case table.
- Execution evidence.
- Defect/fix notes.
- Final test result.

Evidence / Submission Reference

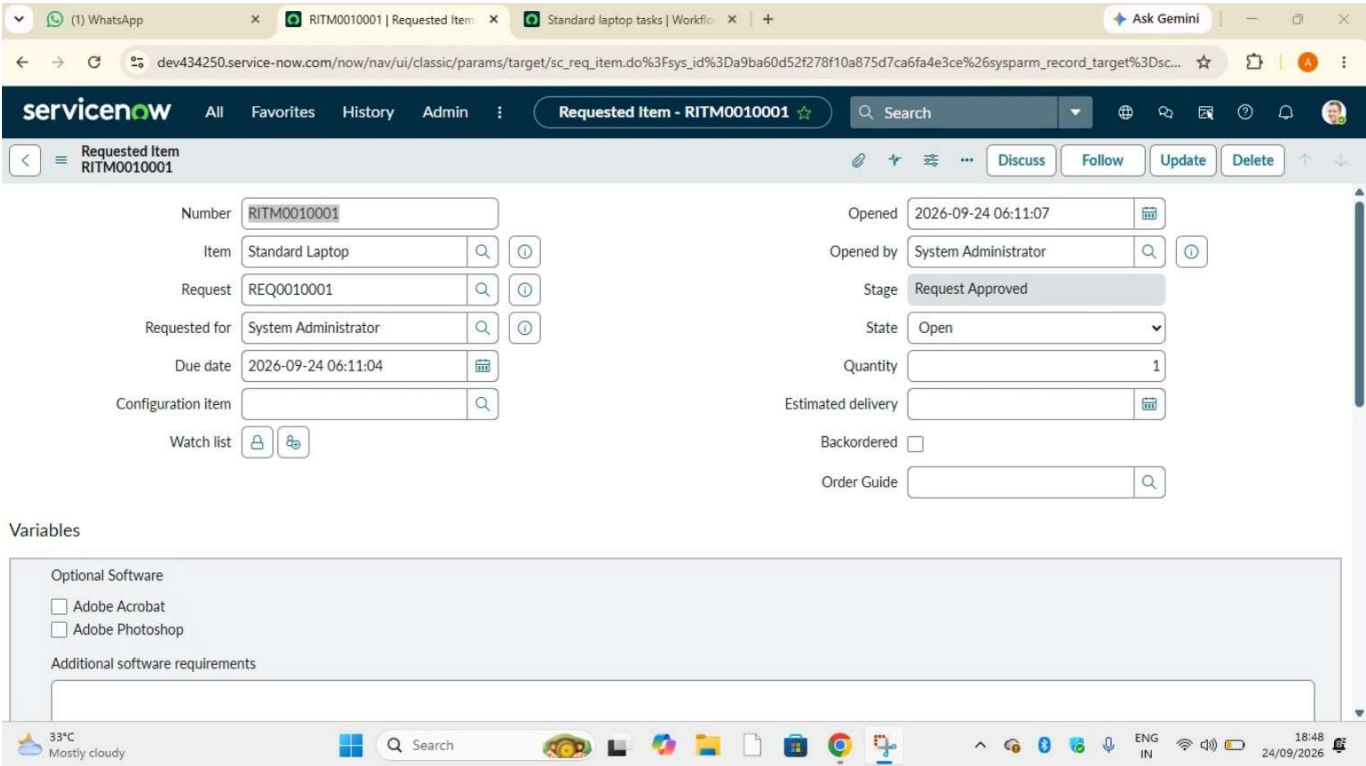
- Screenshots of successful execution.
- Screenshots of alternative/negative testing.
- GitHub folder: 06_Project_Testing/

7.1 Test Case Table

TC ID	Scenario	Input	Expected Result	Actual Result	Status
TC-01	Standard laptop task execution	Service Catalog trigger / standard laptop task	Create Catalog Task action creates a Catalog Task record	SCTASK0010001 created; Approval = Approved; State = Open; Request Item = RITM0010001	PASS

8. Project Documentation Phase

Testing Evidence Screenshot



ServiceNow Catalog Task - SCTASK0010001

Number: SCTASK0010001

Assigned to: [Search]

Configuration item: [Search]

Active: ☒

Approval: Approved

Priority: 4 - Low

State: Open

Request item: RITM0010001

Requested for: System Administrator

Short description: Laptop need to Configured

Description: Laptop need to Configured

Work notes: [Text Area]

Post

Activities: 1

System Administrator

Impact: 3 - Low

Owned by: System Administrator

Field changes: 2026-09-24 06:16:44

Prepare the complete final project record so another person can understand, reproduce and evaluate the work.

Activities / Work to Complete

- Compile the eight required project phases and attach the available implementation evidence.
- Add project overview, objectives, requirements, design and implementation details.
- Added screenshots of the Workflow Studio flow and the resulting Catalog Task record.
- Added the evidence-based testing result for the standard laptop task flow.
- Add GitHub repository link.
- Add limitations and future enhancements.
- Prepare README.md and final project report.

Expected Deliverables

- Final project report.
- README.md.
- Screenshots/evidence: Workflow Studio flow, catalog item configuration, request item, and generated Catalog Task record.
- Public GitHub repository.

Evidence / Submission Reference

- Final report to be uploaded to the Public GitHub repository.
- README screenshot/link.
- GitHub location: 07_Project_Documentation/ (create/upload before final submission).

8.1 README.md – Suggested Content

- Project title and short description.
- Problem statement.
- Objectives.

- Technology used: ServiceNow Flow Designer.
- Project workflow.
- Implementation summary.
- Testing summary.
- Repository structure.
- How to view the project evidence.
- Limitations/future enhancements.

9. Project Demonstration Phase

Demonstrate the working project clearly from input/request to final automated result.

Activities / Work to Complete

- Introduce the problem and project objective.
- Show the ServiceNow environment and relevant module.
- Show the “Standard laptop task” flow in Workflow Studio.
- Explain the Service Catalog trigger and Create Catalog Task action.
- Show the resulting Catalog Task record created for the laptop request.
- Show SCTASK0010001 and its Approved/Open status.
- Show the resulting record/status/notification.
- Show the Public GitHub repository and the uploaded project files/folders.
- Conclude with benefits, limitations and future scope.

Expected Deliverables

- Demo script.
- Working live demonstration.
- GitHub repository demonstration.
- Final screenshots/video if requested.

Evidence / Submission Reference

- Demo screenshots/video.
- GitHub repository link.
- GitHub location: 08_Project_Demonstration/ (create/upload before final submission).

Demonstration Evidence Screenshots

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Standard Laptop	Lenovo - Carbon x1	true		Service Catalog	Hardware	\$1,100.00	Item	2022-11-20 20:46:33
StarTech Mini Display Port to DVI Adapter	PC / Mac Compatible	false		Service Catalog	Peripherals	\$17.23	Item	2022-11-20 20:46:33
StarTech Mini Display Port to VGA Adapter	PC / Mac Compatible	true		Service Catalog	Peripherals	\$24.50	Item	2022-11-20 20:46:33
StarTech USB Mini Hub	4 Port, USB 2.0	true		Service Catalog	Peripherals	\$11.55	Item	2022-11-20 20:46:33
StarTech USB to DVI Adapter	Multi Monitor Adapter	true		Service Catalog	Peripherals	\$49.99	Item	2022-11-20 20:46:33
STM MacBook Air Sleeve	STM Mac Sleeve - S13"	true		Service Catalog	Peripherals	\$24.26	Item	2022-11-20 20:46:33
STM MacBook Pro Sleeve	STM Mac Sleeve - S15"	true		Service Catalog	Peripherals	\$25.78	Item	2022-11-20 20:46:33
Table Index	Do you need to index a database table?	true		Service Catalog	Services	\$0.00	Item	2022-11-20 20:46:33

10. Final Project Summary

The project, “Streamlining IT Procurement: Automatic Standard Laptop Orders With Flow Designer,” implements a standard laptop task workflow in ServiceNow Workflow Studio / Flow Designer. The completed flow is named “Standard laptop task”, uses a Service Catalog trigger, and contains a “Create Catalog Task” action. The implementation evidence shows the flow as Active/Published and a resulting Catalog Task record SCTASK0010001 with Approved approval and Open state.

11. Future Enhancements

- Add approval automation for different laptop categories.
- Integrate inventory/asset availability checks.
- Add automatic vendor or procurement-team communication.
- Add SLA tracking and escalation.
- Add dashboards and reporting for procurement requests.
- Add more laptop models and rule-based selection.
- Improve exception handling and audit tracking.

12. Final Submission Checklist

- ☒ Project is implemented in ServiceNow Flow Designer.
- ☒ GitHub repository is Public.
- ☐ All required project files/documents are uploaded to GitHub before final submission.
- ☐ Eight required phase folders are created in GitHub before final submission.

- ☒ Screenshots/evidence are included in this final document.
- ☒ Testing result documented from the available implementation evidence.
- ☐ README.md is updated with the project details and evidence before final submission.
- ☒ GitHub link is included in the final report.
- ☐ Demonstration is prepared.
- ☒ Final document is checked for names and screenshots.

13. GitHub Repository Link

<https://github.com/ananthivellaichamy05-create/Streamlining-IT-Procurement.git>

Final submission note: The student details and Public GitHub repository link have been added. The technical implementation details are based on the ServiceNow screenshots supplied for this project. Before submitting, upload this final report and the required project evidence/files to the Public GitHub repository, and enter the final submission date.

Submission note: The technical details in this document have been aligned with the ServiceNow screenshots provided for the completed implementation. The screenshots are included as project evidence.